Phase Diagrams

Answer key with explanations

A. Ideas

1. Definitions.

> (a) The single temperature and pressure at which **solid, liquid and gas coexist** in equilibrium. (b) The **end of the liquid–gas line**: the highest temperature and pressure at which liquid and gas can be told apart. (c) The state **beyond the critical point**, with properties between those of a liquid and a gas. (d) Conditions where **two phases coexist** in equilibrium.

B. Water

2. Sketch.

> Temperature on the x-axis, pressure on the y-axis. Solid at upper left, liquid at top centre, gas at lower right. The solid–liquid line rises steeply and slopes **slightly left**. The three lines meet at the **triple point** (0.01 °C, 0.00604 atm). The liquid–gas line ends at the **critical point** (374 °C, 218 atm). A horizontal line at 1 atm crosses the solid–liquid line at **0 °C** and the liquid–gas line at **100 °C**.

3. States of water.

> (a) **liquid** (b) **gas** (c) **solid**: below the triple point, and the pressure (0.001 atm) is above the vapour pressure of ice at −40 °C (0.00013 atm) (d) **supercritical fluid** (above 374 °C and 218 atm)

4. The slope.

> Ice is **less dense** than liquid water (it has a more open, hydrogen-bonded structure). Increasing the pressure favours the denser phase, the **liquid**, so the melting point falls as the pressure rises: the line slopes to the left.

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C. Carbon dioxide

5. States of CO_2.

> (a) **solid** (below −78.5 °C at 1 atm) (b) **gas** (c) **liquid** (above the vapour pressure at 0 °C, below the critical point) (d) **supercritical fluid** (above 31.0 °C and 72.8 atm)

6. Warming CO_2.

> (a) At 1 atm (below the triple-point pressure): the solid **sublimes** directly to gas at −78.5 °C, then the gas warms. (b) At 10 atm (above 5.11 atm): the solid **melts** to liquid (at about −56 °C), the liquid then **boils** to gas at a higher temperature, and the gas warms to 25 °C.

D. Calculations and applications

7. Conversions.

> (a) $72.8\ \cancel{\text{atm}} \times \dfrac{101.325\ \text{kPa}}{1\ \cancel{\text{atm}}} = 7376\ \text{kPa} = \mathbf{7.38 \times 10^3\ kPa}$
>
> (b) $218\ \cancel{\text{atm}} \times \dfrac{101.325\ \text{kPa}}{1\ \cancel{\text{atm}}} = 22\,089\ \text{kPa} = \mathbf{2.21 \times 10^4\ kPa} = \mathbf{22.1\ MPa}$

8. Freeze-drying.

> Below the triple-point pressure of water (0.00604 atm), there is **no liquid region**: the solid borders the gas directly. When the frozen coffee is gently warmed at this low pressure, the ice **sublimes** straight to water vapour, which is pumped away. The coffee is dried without melting, which keeps its structure and flavour.

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